AI-Driven 2D Mesh NoC Router

Project Title

AI-Driven 2D Mesh Network-on-Chip (NoC) Router

Github link

https://github.com/korubilligowtham134/Projectspace-8.0-Team-33

Abstract

Modern System-on-Chip (SoC) architectures contain multiple processing elements such as CPUs, GPUs, memory blocks, AI accelerators, and communication modules. Traditional bus-based communication systems face limitations in scalability, congestion, latency, and power consumption as the number of cores increases. To overcome these limitations, Network-on-Chip (NoC) architecture is used as a scalable and efficient communication framework for multi-core systems.

In this project, an AI-Driven 2D Mesh NoC Router is proposed to improve communication efficiency inside multi-core systems. The router is implemented in SystemVerilog and incorporates a Rule-Based Congestion Detection Algorithm within the AI Controller module. The controller monitors buffer occupancy of neighboring ports and dynamically reroutes packets through alternative paths when congestion is detected, ensuring real-time adaptive routing decisions.

The proposed system is functionally verified using EDA Playground and further analyzed for timing and synthesis using Cadence and Xilinx Vivado respectively. Compared to traditional NoC routers that use fixed static routing, the proposed AI-driven router provides intelligent congestion handling, improved Quality of Service, reduced packet latency, and better overall communication performance.

Introduction

As semiconductor technology advances, modern chips integrate multiple cores and processing units on a single chip, making efficient on-chip communication a critical design challenge. Traditional communication methods such as buses and point-to-point connections are no longer sufficient for large-scale multi-core architectures due to their inherent limitations in handling high traffic volumes, limited scalability, increased wiring complexity, and growing power consumption.

Network-on-Chip (NoC) architecture addresses these challenges by introducing a structured, packet-based communication fabric using routers and links, similar to how internet networks operate. Among the various NoC topologies, the 2D Mesh topology is widely adopted due to its regular structure, simple routing, and ease of scalability. In a 2D mesh, each router is connected to its neighboring routers in four directions — North, South, East, and West — and to a local Processing Element (PE).

However, traditional NoC routers rely on fixed routing algorithms such as XY routing, which forward packets along predetermined paths without considering current traffic conditions. This leads to congestion hotspots, increased packet delay, uneven load distribution, and poor Quality of Service during heavy traffic scenarios.

To address these limitations, this project proposes an AI-Driven 2D Mesh NoC Router that integrates a Rule-Based Congestion Detection Algorithm into the routing process. The AI Controller monitors the buffer occupancy of each neighboring port in real time and intelligently reroutes packets through less congested alternative paths. The entire system is designed and implemented in SystemVerilog, verified functionally on EDA Playground, and analyzed for timing and synthesis using Cadence and Xilinx Vivado tools.

Objectives

Main Objectives

1. Design a 2D Mesh NoC Router.
2. Implement packet-based communication.
3. Reduce congestion inside the network.
4. Improve Quality of Service (QoS).
5. Introduce intelligent routing concepts.
6. Improve scalability for multi-core systems.
7. Reduce packet latency and delay.

Existing System

Traditional NoC Router

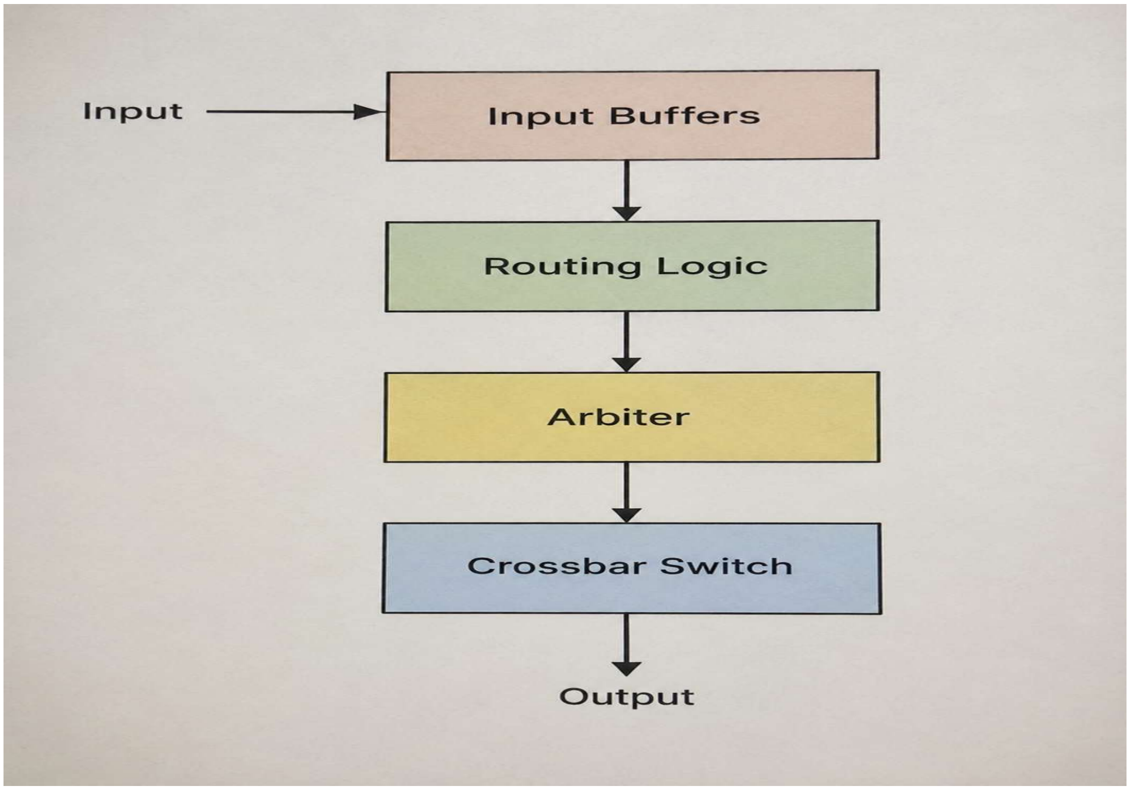
Traditional NoC routers use fixed routing methods. They do not dynamically analyze traffic conditions.

Limitations of Existing System

- Static routing paths
- Congestion during heavy traffic
- Increased packet delay
- Uneven load distribution

- Lower throughput
- Poor QoS handling

Existing Router Architecture



Proposed System

AI-Driven NoC Router

The proposed system introduces intelligent routing and congestion-aware communication in a 2D mesh topology.

The router dynamically selects efficient paths based on network traffic conditions.

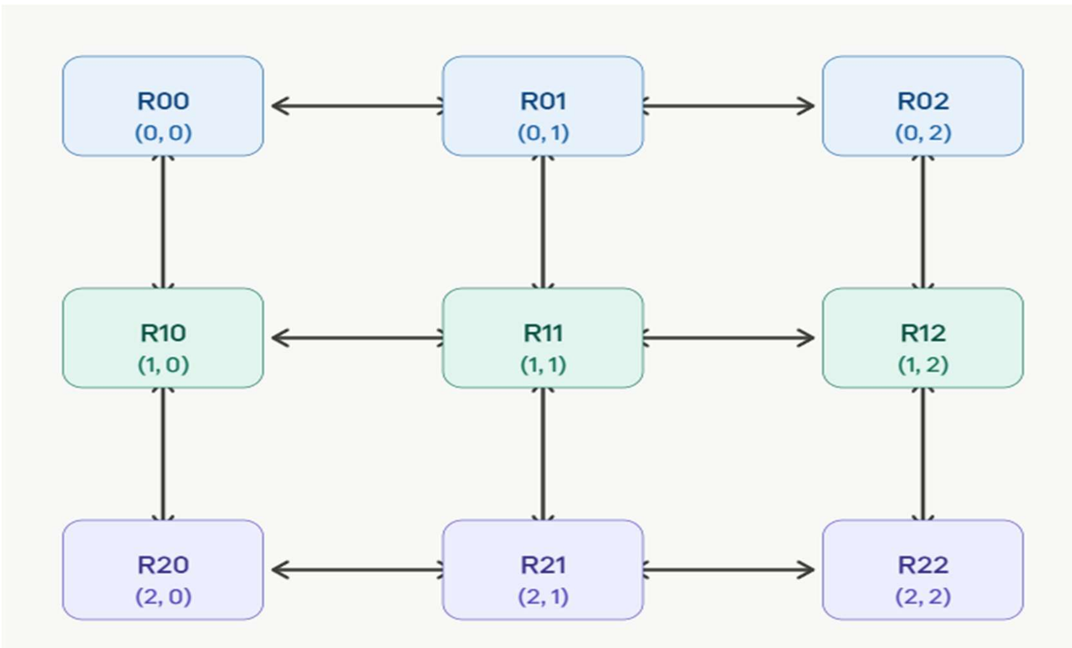
Features of Proposed System

- Intelligent packet routing
- Congestion prediction
- Adaptive path selection
- Better load balancing
- Improved QoS

- Reduced latency
- Higher throughput

2D Mesh Topology

Mesh Architecture Diagram

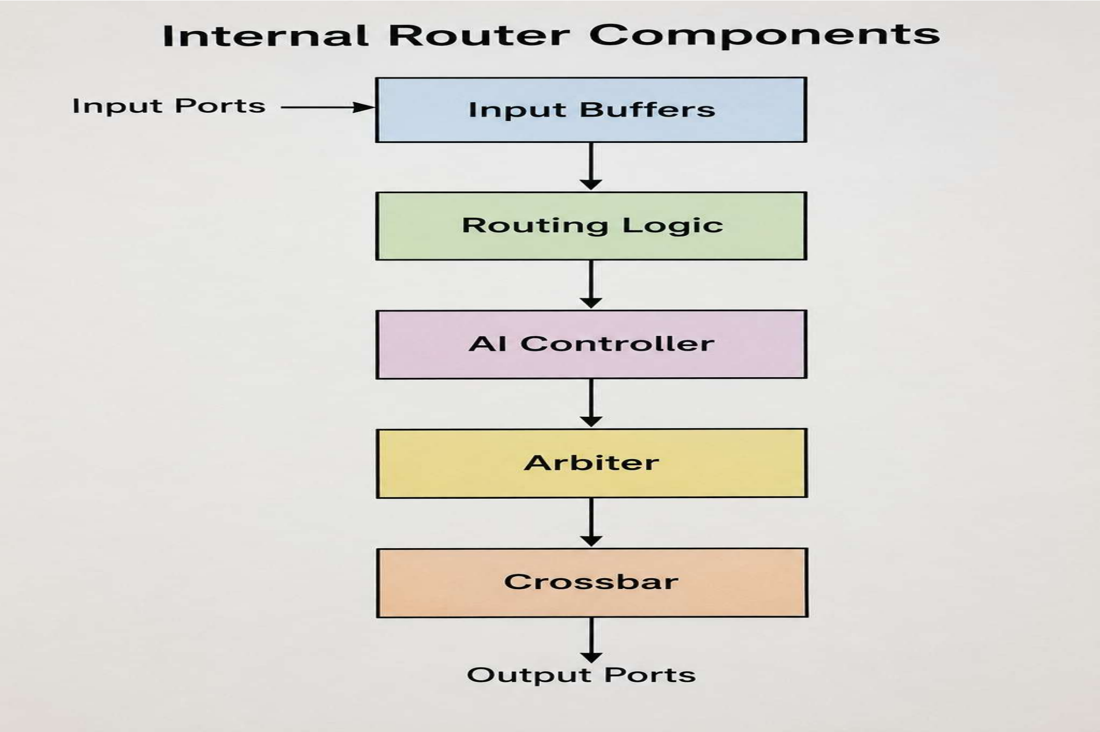


Where:

- R = Router
- Each router connects to a Processing Element (PE)
- Routers communicate in four directions:
    - North
    - South
    - East
    - West

Router Architecture

Internal Router Components



Explanation of Router Components

1. Input Buffer

Input buffers temporarily store incoming packets before processing.

Functions

- Packet storage
- Traffic handling
- Prevent packet loss
- Smooth data flow

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2. Routing Logic

Routing logic determines the next path for packet transmission.

Functions

- Destination checking

- Path selection
- Packet forwarding

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3. AI Controller

The AI Controller in the proposed router implements a Rule-Based Congestion Detection Algorithm in SystemVerilog. This represents a foundational form of artificial intelligence where routing decisions are made based on predefined conditions derived from real-time network behavior. Unlike traditional NoC routers that use fixed static routing regardless of traffic conditions, the proposed AI Controller dynamically adapts its routing decisions based on current network state.

The controller continuously monitors the buffer occupancy of each neighboring port (North, South, East, West) and compares it against a predefined threshold value. If the buffer occupancy of a port exceeds the threshold, that link is flagged as congested and the packet is rerouted through an alternative available path. If the occupancy is below the threshold, the packet follows the standard XY routing path. This decision is made at every router for every incoming packet within a single clock cycle, ensuring real-time congestion avoidance.

This continuous monitoring, evaluation, and adaptive decision-making is what qualifies the system as AI-driven. The approach requires no external training data, introduces minimal hardware overhead, and produces deterministic and synthesizable behavior suitable for real chip implementation.

Advantages

- Intelligent routing
- Better traffic management
- Improved efficiency

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4. Arbiter

The arbiter resolves conflicts when multiple packets request the same output channel.

Functions

- Priority management
- Channel allocation
- Avoid collision

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5. Crossbar Switch

The crossbar switch connects input ports to output ports.

Functions

- Packet switching

- Parallel communication
- Fast transmission

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6. Output Ports

Output ports send packets to neighboring routers or destination processing elements.

Implementation and Methodology Hardware Description Language

The proposed AI-Driven 2D Mesh NoC Router is implemented using System Verilog, a hardware description and verification language that supports both design and testbench development. The design is structured into two primary modules:

Design Under Test (DUT) — describes the actual router hardware including input buffers, routing logic, AI controller, arbiter, and crossbar switch

Testbench (TB) — drives stimulus to the DUT, monitors outputs, and verifies functional correctness of the router under various traffic conditions

AI-Based Improvements

Improvements Over Traditional Router

Traditional NoC	AI-Driven NoC
Static routing	Intelligent routing
Congestion occurs frequently	Congestion-aware routing
Higher delay	Reduced latency
Poor load balancing	Adaptive load balancing
Limited QoS	Improved QoS
Fixed decision making	Dynamic path selection

Congestion Handling Diagram

Traditional Routing

Packet Flow:

```
R1 ---> R2 ---> R3 ---> R4
          (Congestion)
```

Packets wait and delay increases.

AI-Based Routing

Packet Flow:

```
R1 ---> R2 ---> R5 ---> R4
```

Alternative path selected intelligently.

Quality of Service (QoS)

Quality of Service refers to maintaining efficient and reliable communication.

QoS Improvements

- Reduced packet loss
- Reduced latency
- Better bandwidth utilization
- Faster communication
- Efficient packet scheduling

Applications

Applications of AI-Driven NoC Router

- Multi-core processors
- AI accelerators
- GPUs
- Embedded systems
- High-performance computing
- Data centers
- Autonomous systems
- Machine learning hardware

Advantages

Advantages of Proposed System

- Scalable architecture
- Intelligent routing
- Reduced congestion
- Better performance
- Improved packet delivery
- Efficient communication
- Lower latency
- Better bandwidth usage

Future Scope

Future Enhancements

- Machine learning-based prediction models
- Reinforcement learning routing
- Power-aware routing
- Fault-tolerant routing
- Dynamic topology optimization
- Security-aware NoC routing

Conclusion

The AI-Driven 2D Mesh NoC Router improves communication efficiency in modern multi-core systems. By integrating intelligent routing and congestion-aware techniques, the proposed system achieves better traffic management, reduced latency, and improved Quality of Service.

Compared to traditional NoC routers, the proposed architecture provides scalable and adaptive communication suitable for future semiconductor technologies and AI-based computing systems.

References

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